AIDCF

AI Data Centre Facilities Assurance Framework

EVIDENCE MODEL

Version 1.0

A normative evidence architecture for traceable, current, sufficient and reviewable AI infrastructure assurance

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Evidence is the currency of assurance. Assertions, scores and decisions have no independent authority beyond their evidence, scope and review.

Document control

Field	Value
Title	AIDCF Evidence Model v1.0
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Author	Siva Sethumadhavan
Depends on	AIDCF Manifesto; Core Conceptual Model; AIDCF-003; AIDCF-004; AIDCF-005; AIDCF-006; AIDCF-007
Purpose	Define evidence objects, classes, metadata, provenance, integrity, quality, freshness, coverage, sufficiency, confidence, conflict, retention and decision use.
Applies to	All AIDCF controls, assertions, findings, critical gates, tests, scores, maturity determinations, reports, decisions and compatible tooling.
Downstream	AIDCF-009 Assessment Methodology; scoring, maturity, reporting and future scheme artifacts.
Safety/privacy boundary	Evidence collection shall be authorised, proportionate, lawful, minimally intrusive and protected. This standard does not authorise hazardous testing or unrestricted data collection.

Executive thesis

AIDCF assurance depends on whether a precise assertion is supported by evidence that is relevant, authentic, sufficiently complete, current for the represented state, traceable to its source, protected against inappropriate alteration, and reviewed by a competent accountable party. Evidence is not graded once for all purposes. Its usefulness is evaluated against the claim, Capability Instance, Service Outcome, boundary, operating state, period and decision question.

Canonical evidence chain

Decision Question → Assertion → Evidence Requirement → Evidence Object → Provenance and Integrity → Relevance and Coverage → Quality and Freshness → Sufficiency → Confidence → Conflict and UNKNOWN → Finding / Critical Gate → Bounded Assurance Decision

Key outputs

- Canonical EvidenceObject, EvidencePackage and EvidencePortfolio structures.
- Seven evidence classes and controlled subtypes.
- Six-dimensional evidence quality model.
- Freshness, retention and chain-of-custody rules.
- Assertion-specific sufficiency and confidence model.
- Evidence-to-control and evidence-to-critical-gate mappings.
- Conflict, derived evidence, automation and UNKNOWN rules.
- Machine-readable ontology, schema and validation expectations.

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Part I - Authority, scope and conformance

1. Purpose

This standard operationalises the evidence commitments in AIDCF-006 and the minimum evidence expectations in AIDCF-007. It defines a common evidence language suitable for human assessments, structured workpapers, graph implementations and future assurance schemes.

2. Normative requirements

EVM-REQ-001 Every material assertion shall link to supporting, challenging or qualifying Evidence Objects, or explicit UNKNOWN.

EVM-REQ-002 Evidence shall retain identity, source, collection, scope, time, integrity, custody, classification and review metadata.

EVM-REQ-003 Evidence quality and sufficiency shall be evaluated against a precise assertion; no evidence item has a universal grade.

EVM-REQ-004 Design, implementation, operation and effectiveness claims shall use evidence appropriate to the claim type.

EVM-REQ-005 Evidence conflict, staleness, inaccessible sources and coverage gaps shall remain visible.

EVM-REQ-006 UNKNOWN shall not be transformed into pass, zero, false, Not Applicable or accepted risk.

EVM-REQ-007 Critical-gate evidence shall meet the declared evidence floor or the gate shall remain UNKNOWN or Fail according to the criterion.

EVM-REQ-008 Derived and AI-generated evidence shall retain source lineage, transformation logic and candidate review state.

EVM-REQ-009 Evidence collection and testing shall be authorised, competent, proportionate, lawful and safe.

EVM-REQ-010 Retention and deletion shall preserve legal, contractual, audit and decision-lineage obligations.

EVM-REQ-011 Evidence reuse across controls or methods shall require assertion-specific relevance and scope evaluation.

EVM-REQ-012 A score, maturity level or certificate shall not increase the evidential strength of its underlying sources.

3. Scope

- Evidence created or collected for all 360 AIDCF-007 controls.
- Evidence supporting infrastructure, dependency, failure, safeguard, recovery and graph assertions.
- Evidence needed to evaluate the 15 baseline critical gates.
- Evidence used by AIDCF and AI Trust Graph integrated decision packages.
- Human, machine, documentary, observational, test, operational and derived evidence.

4. Conformance levels

Level	Name	Minimum expectation
E0	Vocabulary aligned	Uses canonical evidence terms and source ownership.
E1	Metadata complete	Implements mandatory EvidenceObject fields.
E2	Provenance controlled	Adds source, collection, integrity, custody and classification.
E3	Assertion linked	Evaluates relevance, quality, freshness and coverage for precise assertions.
E4	Decision ready	Supports sufficiency, confidence, gates, conflict, UNKNOWN and bounded decisions.
E5	Machine conformant	Passes schemas, validation shapes, versioning and audit tests.

Part II - Evidence doctrine and mandatory distinctions

5. Foundational doctrines

Doctrine	Rule
Evidence before conclusion	No positive assurance conclusion without affirmative support.
Assertion specificity	Evidence supports or challenges a precise proposition, not an object in the abstract.
Reality over intent	Observed and tested evidence takes precedence where design intent conflicts.
Bounded use	Evidence applies only to stated scope, period, state, conditions and method.
Independent review	Collection, interpretation and decision authority remain distinguishable.
Minimum false precision	Scores express structured judgment but do not manufacture certainty.
Preserved uncertainty	Missing, inaccessible, stale, conflicting or weak evidence remains visible.
Non-inheritance	Third-party certification or prior review is evidence, not automatic assurance.

6. Mandatory distinctions

Do not collapse	Reason
Evidence Object / Evidence Package	One item differs from a governed collection.
Source / Collector / Custodian / Reviewer	Origin, acquisition, storage and judgment are different roles.
Observation time / collection time / record time	The represented state may differ from acquisition or repository entry.
Authenticity / integrity	Genuine origin differs from freedom from unauthorised alteration.
Reliability / relevance	A reliable source may not support the claim.
Coverage / volume	Many records may still cover a narrow population or state.
Quality / sufficiency	Strong evidence can still be insufficient in quantity or diversity.
Evidence grade / confidence	Evidence characteristics differ from reviewer confidence in an assertion.
Stale / contradicting	Old evidence differs from current disagreement.
Independent / external	External does not necessarily mean impartial, competent or verified.
Test result / effectiveness conclusion	A test output requires scope and interpretation.
Derived evidence / source evidence	Analysis does not replace its lineage sources.

Part III - Evidence metamodel and object hierarchy

7. Core evidence entities

Entity	Definition
EvidenceObject	Smallest independently identified evidence item used to support, challenge or qualify an assertion.
EvidenceSet	A purpose-specific selection of Evidence Objects used together for one assertion or test.
EvidencePackage	A governed collection addressing a control, function, system, scenario or gate.
EvidencePortfolio	A capability-level collection of Evidence Packages and decision records.
EvidenceSource	Person, system, device, provider, repository or process from which evidence originates.
CollectionActivity	Authorised action that acquires or observes evidence.
TransformationActivity	Declared process deriving one evidence object from others.
EvidenceRequirement	Claim-specific statement of required types, quality, coverage, age and independence.
EvidenceAssessment	Recorded evaluation of evidence against a requirement.
EvidenceManifest	Machine-readable inventory with identifiers, hashes, locations and status.
CustodyEvent	Transfer, access, export, alteration, redaction or retention action.
EvidenceConflict	Structured disagreement between evidence-supported assertions.

8. Hierarchy

EvidencePortfolio → EvidencePackage → EvidenceSet → EvidenceObject → Source and Collection Activity → Assertion / Control / Gate / Decision

9. EvidenceObject lifecycle states

State	Meaning
EXPECTED	Evidence requirement identified but object not yet collected.
REQUESTED	Collection request issued.
COLLECTED	Object acquired but not quality reviewed.
VALIDATED	Identity, provenance, integrity and readability reviewed.
ACCEPTED	Approved for one or more stated assurance uses.
CHALLENGED	Material concern or contradiction recorded.
STALE	No longer sufficiently representative.
RESTRICTED	Use limited by security, privacy, contract or legal hold.
SUPERSEDED	Replaced by a later object while history remains.
REJECTED	Not suitable for the stated use; rationale retained.
RETIRED	No longer current but retained according to policy.
DESTROYED	Disposed under authorised retention process; destruction record remains.

10. Evidence Package minimum structure

Component	Requirement
Package ID and purpose	Stable identity and decision/claim addressed.

Component	Requirement
Subject and boundary	Capability, control, system, gate, state and period.
Requirement profile	Required evidence classes, quality, coverage, freshness and independence.
Manifest	Complete list of included, missing, substituted and restricted items.
Assessments	Per-object quality and package sufficiency results.
Conflicts and UNKNOWN	Unresolved contradictions, gaps and decision effects.
Review and approval	Collector, analyst, reviewer and decision roles.
Version and custody	Immutable release, hash, access and supersession lineage.

Part IV - Evidence taxonomy and subtype catalogue

Class	Name	Representative subtypes
E1	Documentary and Declarative	policy; standard; procedure; runbook; design basis; architecture; drawing; contract; attestation; inventory
E2	Configuration and Technical State	device configuration; firmware inventory; cluster manifest; access policy; protection setting; topology export; configuration snapshot
E3	Operational and Observational	telemetry; log; alarm; event; ticket; work order; capacity metric; performance metric; physical observation; inspection record
E4	Validation and Test	commissioning; load test; failover test; recovery test; simulation; scenario exercise; functional test; calibration; witnessed demonstration
E5	Independent and Third-Party	independent engineering review; audit; provider assurance; witnessed test; laboratory result; accredited certificate; regulatory inspection
E6	Human and Experiential	interview; workshop; walkthrough; operator demonstration; expert judgment; shift observation; competency evidence
E7	Derived and Analytical	trend; correlation; capacity analysis; failure analysis; dependency path; reconciled inventory; score; model output; AI-assisted extraction

E1 Documentary and Declarative

Best suited to intent, requirements, authority, design and declared process. It shall not alone prove current implementation or effectiveness.

Subtype family	Examples	Primary claim supported
Documentary and Declarative	policy, standard, procedure, runbook, design basis, architecture, drawing, contract, attestation, inventory	intent, governance and approved design

E2 Configuration and Technical State

Best suited to implementation and technical-state assertions. It requires provenance, extraction context and reconciliation to the current architecture.

Subtype family	Examples	Primary claim supported
Configuration and Technical State	device configuration, firmware inventory, cluster manifest, access policy, protection setting, topology export, configuration snapshot	implementation and technical state

E3 Operational and Observational

Best suited to operation over a stated period. Coverage, completeness, clock integrity, gaps and collection health shall be assessed.

Subtype family	Examples	Primary claim supported
Operational and Observational	telemetry, log, alarm, event, ticket, work order, capacity metric, performance metric, physical observation, inspection record	operation, performance and incidents

E4 Validation and Test

Best suited to effectiveness and resilience under stated conditions. Test authorization, initial state, method, expected result and anomalies shall be retained.

Subtype family	Examples	Primary claim supported
Validation and Test	commissioning, load test, failover test, recovery test, simulation, scenario exercise, functional test, calibration, witnessed demonstration	validation and effectiveness

E5 Independent and Third-Party

May strengthen independence but shall be evaluated for scope, competence, method, age, exclusions and transferability.

Subtype family	Examples	Primary claim supported
Independent and Third-Party	independent engineering review, audit, provider assurance, witnessed test, laboratory result, accredited certificate, regulatory inspection	independent corroboration

E6 Human and Experiential

Useful for context, operating practice and explanation. Corroboration is required for material technical-effectiveness assertions.

Subtype family	Examples	Primary claim supported
Human and Experiential	interview, workshop, walkthrough, operator demonstration, expert judgment, shift observation, competency evidence	practice, competence and context

E7 Derived and Analytical

Useful for synthesis and decision support. Derived outputs never exceed the quality, scope and lineage of their sources.

Subtype family	Examples	Primary claim supported
Derived and Analytical	trend, correlation, capacity analysis, failure analysis, dependency path, reconciled inventory, score, model output, AI-assisted extraction	analysis and relationship synthesis

Part V - Evidence metadata standard

11. Mandatory metadata

Field	Requirement
evidenceId	Immutable unique identifier.
title	Concise human-readable name.
description	What the item contains and why it exists.
evidenceClass/subtype	Controlled E1-E7 classification.
sourceId/sourceType	Originating person, system, provider or process.
owner/custodian	Accountable owner and storage custodian.
collector/collectionMethod	Who or what collected it, how and under which authority.
observationTime	When the represented condition existed.
collectionTime	When the evidence was acquired.
recordTime	When entered into the repository.
effectivePeriod	Period for which the evidence is applicable.
subjectRefs	Capabilities, assets, controls, gates, assertions or paths addressed.
scope	Locations, systems, tenant, population and states covered.
architecture/configurationVersion	Baseline represented.
integrityMethod/hash	Integrity protection and digest where applicable.
format/version	File, record, schema and tool versions.
classification	Security/privacy/contractual handling category.
retentionRule	Retention period, trigger and legal hold.
qualityAssessment	Dimension ratings and rationale.
reviewState/reviewer	Candidate, accepted, challenged, rejected, stale or superseded.
limitations	Known gaps, exclusions, sampling or interpretation limits.
relatedEvidence	Sources, derivatives, conflicts, replacements and package membership.

12. Conditional metadata

- Test evidence: initial state, expected result, procedure, authorization, abort criteria, instrumentation, anomalies and witnesses.
- Telemetry/log evidence: source clock, timezone, sampling interval, collection health, dropped data and query/filter.
- Configuration evidence: extraction command or interface, privilege used, configuration scope, redactions and target state.
- Human evidence: role, competence, interview protocol, date, corroboration and consent/recording status.
- External evidence: issuer, competence/accreditation where relevant, scope, validity, exclusions and permitted reliance.
- Derived evidence: source IDs, transformation code/query/version, assumptions, quality propagation and reviewer.

Part VI - Provenance, integrity and chain of custody

13. Provenance model

Provenance object	Minimum content
Entity	Evidence object or source record.
Activity	Collection, observation, transformation, review or custody event.
Agent	Person, system, provider or organisation responsible.

Provenance object	Minimum content
wasGeneratedBy	Activity that produced the object.
wasDerivedFrom	Source evidence and transformations.
wasAttributedTo	Responsible source or owner.
wasAssociatedWith	Collector, reviewer or tool involved.
used	Source objects consumed by a transformation.
invalidatedAtTime	When evidence became stale, superseded, withdrawn or destroyed.

14. Integrity levels

Level	Name	Anchor
I0	Unknown	Integrity method absent or unverified.
I1	Basic controlled	Repository permissions and version history.
I2	Verified copy	Source comparison or controlled export metadata.
I3	Hashed	Cryptographic digest recorded in manifest.
I4	Signed / trusted timestamp	Digital signature or trusted timestamp validated.
I5	Tamper-evident custody	End-to-end signed custody, immutable storage or equivalent control.

15. Chain-of-custody events

- Creation or acquisition.
- Transfer between collectors, custodians or repositories.
- Read/access for review.
- Transformation, parsing, redaction or export.
- Integrity verification.
- Challenge, correction or supersession.
- Legal hold, restriction, retention extension or authorised destruction.

EVM-REQ-015 Every material transformation shall produce a new Evidence Object linked to source evidence; original evidence shall not be silently overwritten.

EVM-REQ-016 Redaction shall preserve an unredacted controlled source where lawful and shall record the basis, method and affected content.

Part VII - Evidence quality dimensions and grading

16. Quality dimensions

Code	Dimension	Question
P	Provenance	Can origin, authority and collection lineage be established?
F	Freshness	Is the item sufficiently current and representative?
C	Coverage	Does it cover the required population, period, state and boundary?
R	Reliability	Is the method/source repeatable, controlled and corroborated?
T	Traceability	Can it be linked end to end to subject, assertion, control and decision?
I	Integrity	Is it protected against undetected alteration and custody loss?

17. Rating scale

P rating anchors

Rating	Anchor
P0	Unknown origin
P1	Identified but uncontrolled source
P2	Accountable internal source
P3	Controlled authoritative source
P4	Corroborated authoritative source
P5	Independently verified origin and lineage

F rating anchors

Rating	Anchor
F0	Unknown age or state
F1	Obsolete for represented state
F2	Stale or change-sensitive
F3	Acceptable for decision
F4	Current and aligned to baseline
F5	Near-real-time or event-current with healthy collection

C rating anchors

Rating	Anchor
C0	Unknown coverage
C1	Anecdotal or isolated
C2	Limited sample/state
C3	Representative core scope
C4	Broad scope and relevant states

Rating	Anchor
C5	Comprehensive declared population, period and states

R rating anchors

Rating	Anchor
R0	Unassessed
R1	Weak/uncontrolled method
R2	Plausible but lightly corroborated
R3	Controlled repeatable method
R4	Strong and independently corroborated
R5	Validated method and independent verification

T rating anchors

Rating	Anchor
T0	Unlinked
T1	Manual contextual link
T2	Linked to subject or control
T3	Linked to assertion, subject and period
T4	End-to-end to finding and gate
T5	End-to-end to decision with immutable lineage

I rating anchors

Rating	Anchor
I0	Unknown integrity
I1	Basic repository controls
I2	Verified source copy
I3	Hash verified
I4	Signature/timestamp verified
I5	Tamper-evident end-to-end custody

18. Evidence quality vector

Preferred expression: [Class]-P#-F#-C#-R#-T#-I#. Example: E4-P4-F4-C3-R4-T4-I3. The vector shall remain visible. A single averaged number may be used only as a secondary sorting aid and shall not replace the dimensions.

19. Evidence quality bands

Band	Rule	Permitted use
Q0 Unusable	Any critical dimension P0, R0, T0 or I0 without accepted limitation.	Context only; no positive assertion.
Q1 Weak	Multiple dimensions 0-1 or unsupported source.	Candidate/UNKNOWN; remediation priority.
Q2 Limited	Core provenance established but limited coverage/currentness/corroboration.	Descriptive or limited implementation claim.

Band	Rule	Permitted use
Q3 Adequate	All claim-critical dimensions at least 3.	Design/implementation claim where appropriate.
Q4 Strong	Most dimensions 4+, with no claim-critical dimension below 3.	Operating effectiveness with sufficient package coverage.
Q5 Highly defensible	Independent/corroborated, representative, current and traceable with strong integrity.	High-consequence gate or resilience claim, subject to method and sufficiency.

Part VIII - Freshness, validity, retention and legal hold

20. Freshness rule

Freshness is risk- and change-based. The windows below are default review triggers, not universal expiry. A material change invalidates representativeness even when the nominal window has not elapsed.

Evidence category	Default currency basis	Immediate invalidation/review triggers
Live telemetry and active alarms	Current collection health; decision-specific lookback	Collection outage, sensor/configuration change, incident.
Event and security logs	Usually current period and sufficient lookback for decision	Clock/source/query change, incident, retention gap.
Configuration snapshots	At assessment and after material change	Firmware, policy, topology, identity or setpoint change.
Capacity/performance metrics	Current demand cycle and peak/seasonal state	Expansion, workload shift, throttling or demand change.
Maintenance/calibration records	Current maintenance cycle	Overdue work, failure, deferral or scope change.
Commissioning and validation tests	Until material change, adverse event or review trigger	Architecture, load, firmware, procedure or provider change.
Recovery and continuity exercises	Risk-based periodic cycle and after material change	Failure, failed exercise, new dependency or recovery design change.
Architecture/design documents	Until superseded or material change	As-built discrepancy, expansion or reconfiguration.
Contracts/provider attestations	Until expiry, renewal or service change	Provider, SLA, subprocessor or jurisdiction change.
Interviews and walkthroughs	Point-in-time context	Role/process change or contradictory technical evidence.

21. Retention classes

Class	Purpose	Rule
R0 Transient	Temporary extraction or working copy.	Delete under controlled process after verified ingestion unless needed for decision traceability.
R1 Operational	Routine monitoring and work records.	Retain for operational, contractual and investigation needs.
R2 Assurance	Assessment evidence and workpapers.	Retain through decision validity plus defined challenge/reassessment period.
R3 Critical decision	Gate, resilience and high-consequence evidence.	Longer risk-based retention with immutable lineage.
R4 Legal/regulatory hold	Evidence subject to preservation obligation.	No alteration/destruction until authorised release.
R5 Canonical methodology	Schemas, mappings, approved examples and release artifacts.	Versioned permanent archive.

EVM-REQ-021 Retention periods shall be defined by applicable legal, contractual, regulatory, security, privacy and operational requirements; this standard does not prescribe universal years.

Part IX - Coverage, sampling and representativeness

22. Coverage dimensions

Dimension	Coverage question
Entity	Which assets, systems, providers and interfaces are included?
Population	What proportion and which strata of the population are represented?
Location	Which sites, halls, zones, racks and jurisdictions?
Time	Which period, season, demand cycle and incident window?
Operating state	Normal, degraded, maintenance, emergency, recovery and transition?
Scenario	Which threat, hazard, failure, common-mode and recovery paths?
Tenant/workload	Which tenancy, cluster, workload or service outcomes?
Control behavior	Design, implementation, operation, monitoring, validation and recovery?
Dependency	Which upstream, downstream, shared and external dependencies?

23. Sampling principles

- Sampling shall be risk-based, reproducible and documented.
- The population, strata, selection method, sample size rationale and exclusions shall be recorded.
- Judgmental sampling may target critical or anomalous items but shall not be represented as statistical coverage.
- Random or systematic sampling requires a trustworthy population frame.
- A sampled pass shall not be extrapolated beyond the represented population and conditions.
- Exceptions and failed samples shall influence expansion and conclusion.

24. Coverage rating

Level	Meaning
C0	Population or scope unknown.
C1	Single example or anecdotal evidence.
C2	Limited sample or one state.
C3	Representative core population and expected state.
C4	Broad population and material degraded/maintenance states.
C5	Comprehensive declared population, periods and applicable states, subject to practical limits.

Part X - Evidence sufficiency and confidence

25. Sufficiency model

Sufficiency is an assertion-specific judgment based on whether the combined evidence package meets required type diversity, quality floors, coverage, freshness, independence, conflict resolution and state/scenario representation. It shall not be calculated only from evidence count.

Level	Name	Meaning
S0	None	No usable evidence.
S1	Insufficient	Some evidence exists but cannot support the claim.
S2	Minimum descriptive	Supports a bounded description or design intent.
S3	Reasonable	Supports design or implementation within declared scope.

Level	Name	Meaning
S4	Strong	Supports operating effectiveness for representative conditions.
S5	Highly defensible	Supports high-consequence gate/resilience claim with diverse, current, traceable and appropriately independent evidence.

26. Sufficiency decision factors

Factor	Assessment
Required class diversity	Are design, configuration, operation and validation types present as required?
Quality floor	Does each mandatory item meet claim-specific minimums?
Complementarity	Do items address different weaknesses rather than duplicate one source?
Independence	Are critical claims corroborated by an appropriately separate source/method?
Coverage	Does the package represent population, state, time and scenario?
Conflict	Are material contradictions resolved or bounded?
UNKNOWN	Are missing elements identified with decision effect?
Critical gate	Does the package meet the declared non-compensating floor?

27. Confidence states

State	Meaning
CONF-0 UNKNOWN	No defensible confidence state.
CONF-1 Low	Weak/limited evidence; high sensitivity to assumptions.
CONF-2 Moderate	Reasonable support with material limitations.
CONF-3 High	Strong support with no unresolved material conflict within scope.
CONF-4 Very High	Highly defensible, independently corroborated and representative support for the bounded claim.
CONFLICTING	Material evidence supports incompatible propositions.
STALE	Previously supported but no longer sufficiently current.

EVM-REQ-027 Confidence shall be recorded with rationale and shall not be derived from a hidden formula.

EVM-REQ-028 A confidence state shall not exceed the ceiling imposed by the weakest mandatory evidence requirement or critical UNKNOWN.

Part XI - Evidence-to-assertion and finding model

28. Assertion structure

Field	Requirement
assertionId	Stable versioned identity.
proposition	Subject-predicate-object or equivalent precise statement.
claimType	Intent, design, implementation, operation, effectiveness, resilience, absence, compliance mapping or decision.
scope/conditions	Capability, boundary, state, configuration, load, period and assumptions.
evidenceRequirement	Required classes, quality floors, coverage and independence.
supportingEvidence	Evidence Objects supporting proposition.
challengingEvidence	Evidence Objects contradicting or weakening proposition.
qualifyingEvidence	Evidence limiting scope or interpretation.
assessment	Quality, sufficiency, conflict and UNKNOWN evaluation.
confidence/review	Controlled state, reviewer and date.
decisionUse	Findings, gate or decision that relies upon assertion.

29. Claim-type evidence floors

Claim type	Minimum pattern
Intent/governance	E1 plus accountable approval and current applicability.
Design suitability	E1 design evidence plus competent technical analysis; E5 where required by authority or consequence.
Implemented	E2 configuration/as-built evidence plus E3 or E6 observation; reconcile to design.
Operating	E3 over representative period plus current E2 and exception/incident review.
Effective	E4 representative validation plus E2/E3 supporting state and operating evidence.
Resilient/recoverable	E4 scenario/recovery evidence plus cross-domain E2/E3 and dependency coverage.
Negative/absence claim	Documented discovery coverage, method sensitivity and residual UNKNOWN.
Independent assurance	E5 scope and method plus access to underlying evidence where needed.
Derived decision	E7 with complete source lineage and human review.

30. Finding model

Finding state	Evidence rule
Observation	Evidence describes a condition without determining control result.
Conforming	Affirmative evidence meets requirement and sufficiency floor.
Partial	Some requirement elements or coverage are supported.
Nonconforming	Evidence demonstrates requirement not met.
UNKNOWN	Evidence insufficient, inaccessible, stale or unresolved.
Conflicting	Material sources disagree.
Not Applicable	Approved evidence-based applicability rationale.
Suspended	Change or expiry invalidates reliance on prior evidence.

Part XII - Evidence-to-control mapping

31. Control evidence profile

Every AIDCF-007 control shall have an Evidence Requirement Profile. The profile is normative for evidence types and floors but may be strengthened by capability profile, criticality, scenario or decision need.

Field	Meaning
controlId	AIDCF-007 control.
claimType	Design, implementation, operation, effectiveness or resilience.
requiredClasses	Mandatory and optional E1-E7 classes.
qualityFloor	Minimum P/F/C/R/T/I dimensions.
sufficiencyFloor	S0-S5.
stateCoverage	Operating states to be represented.
periodCoverage	Required period or risk-based window.
independence	Required source/method separation.
mandatoryItems	Non-substitutable evidence.
acceptableAlternatives	Equivalent evidence with rationale.
unknownEffect	Decision or gate consequence.
reassessmentTrigger	Evidence invalidation conditions.

32. Control profile tiers

Tier	Use	Minimum evidence pattern
CP-1 Foundational	Governance, inventory, design intent.	E1 plus accountable source; S2/S3.
CP-2 Technical state	Configuration, topology and implementation.	E1+E2 with observation/reconciliation; S3.
CP-3 Operational	Operation, monitoring and maintenance.	E2+E3 across representative period; S3/S4.
CP-4 Effectiveness	Controls tested under defined conditions.	E2+E3+E4; S4.
CP-5 Critical/resilience	Critical gate, compound scenario or high-consequence outcome.	Diverse E1-E4 and appropriate E5; S5 or profile-defined strong floor.

Part XIII - Critical-gate evidence profiles

Gate	Name	Mandatory evidence profile	Classes	Floor	Decision effect
CG-01	Capability and outcome definition	Capability register; approved outcomes; criticality and tolerance analysis; architecture linkage	E1+E7	S3	UNKNOWN capability/outcome blocks positive capability assurance.
CG-02	Boundary and architecture integrity	Approved boundary; current architecture; as-built/configuration reconciliation; exclusions and external dependencies	E1+E2+E6	S3	Unclear material boundary blocks final conclusion.
CG-03	Critical dependency visibility	Dependency graph; source/route/provider evidence; common-mode analysis; ownership	E1+E2+E7	S4	Critical hidden dependency remains UNKNOWN or Fail if demonstrated absent.
CG-04	Life safety and hazardous energy	Authorised engineering/safety records; inspection; permits; tests; unresolved hazard status	E1+E4+E5	S4	Unsafe unresolved condition blocks relevant assurance/testing.
CG-05	Resilient electrical supply	SLD; protection/capacity study; UPS/generator tests; fuel; maintenance; EPMS/incident evidence	E1+E2+E3+E4	S5	Missing critical continuity evidence → UNKNOWN; demonstrated inability → Fail.
CG-06	Thermal continuity	Thermal design/model; P&ID; capacity/flow telemetry; leak/transfer/failover tests; maintenance	E1+E2+E3+E4	S5	Unverified critical heat-removal path → UNKNOWN; demonstrated shortfall → Fail.
CG-07	Critical control-system integrity	BMS/EPMS/controller architecture; configuration; identity/authority; telemetry; fail-safe tests	E1+E2+E3+E4	S4	Uncontrolled or unverifiable critical automation limits/blocks assurance.
CG-08	Compute, storage and network functional integrity	Current topology/configuration; capacity/performance telemetry; error data; failover/recovery tests	E2+E3+E4	S4	Unverified critical infrastructure function caps operating/resilience assurance.
CG-09	Failure containment	Fault-domain model; isolation design; scenario analysis; containment tests/incidents	E1+E2+E4+E7	S4	Absent/failed containment may be Fail; untested critical boundary is UNKNOWN.
CG-10	Recovery readiness	Recovery plan; prerequisites; configuration/data sources; exercises/tests; defects and signoff	E1+E2+E3+E4	S5	Plan without representative test supports design only, not recoverability.
CG-11	Evidence sufficiency	Evidence requirements; manifest; quality/sufficiency assessments; conflicts and limitations	E1+E7	S4	Failure to meet evidence floor blocks the dependent claim.
CG-12	Critical UNKNOWN	UNKNOWN register; consequence; resolution ownership; evidence requests; decision ceiling	E1+E7	S3	Unresolved critical UNKNOWN caps, conditions, suspends or blocks assurance.
CG-13	Responsibility and authority	RACI; delegations; owner/operator/provider clauses; escalation; decision authority	E1+E6	S3	Unowned critical function/dependency blocks accountable disposition.
CG-14	Change control	Baseline; change request; impact analysis; test/rollback; approval; post-change validation	E1+E2+E3+E4	S4	Material uncontrolled change suspends affected evidence/decision.
CG-15	Integrated digital authority	Identity and privilege records; reachability; control-plane interfaces; session logs; action/rollback tests	E2+E3+E4+E7	S4	Unverified consequential authority remains critical UNKNOWN.

EVM-REQ-033 Evidence for a critical gate shall not be substituted below its mandatory type or quality floor merely to produce a complete scorecard.

Part XIV - Conflict, contradiction and UNKNOWN

33. Evidence relationship states

State	Meaning
SUPPORTS	Evidence increases support for a precise assertion.
CHALLENGES	Evidence weakens or contradicts assertion.
QUALIFIES	Evidence narrows scope or conditions.
INCOMPLETE	Evidence covers only part of requirement.
STALE	Evidence no longer represents current state.
INACCESSIBLE	Evidence is believed to exist but cannot be reviewed.
UNVERIFIED	Origin, integrity or method is not sufficiently established.
SUPERSEDES	New object replaces current use while preserving history.
DUPLICATES	Same underlying source is represented more than once.
CONFLICTS_WITH	Two supported propositions are materially incompatible.

34. Conflict workflow

Detect → register separate assertions → preserve both evidence lineages → assess scope/time/state differences → seek corroboration → competent review → resolve, bound or escalate → update decision while preserving history

35. UNKNOWN categories

Code	Category	Definition
U1	Missing	Expected evidence not produced.
U2	Inaccessible	Evidence exists but access is restricted or unavailable.
U3	Stale	Evidence does not represent current state.
U4	Conflicting	Material sources disagree.
U5	Coverage gap	Required population, state, period or scenario not represented.
U6	Source unverified	Origin/authority cannot be established.
U7	Integrity uncertain	Alteration or custody cannot be ruled out.
U8	Method uncertain	Collection/test method insufficiently defined.
U9	Interpretation unresolved	Competent reviewers cannot yet determine meaning.
U10	Dependency unknown	Evidence cannot establish a material external/shared dependency.

EVM-REQ-035 Each UNKNOWN shall identify subject, missing attribute, reason, potential consequence, evidence needed, owner if known, status and decision/gate impact.

Part XV - Derived, automated and AI-generated evidence

36. Derived evidence rules

- Retain every source Evidence ID and the transformation method.
- Record query, script, model, parameters, assumptions and version.
- Do not assign a quality dimension higher than source support without independent corroboration relevant to that dimension.
- Distinguish deterministic transformation from interpretive inference.
- Preserve failed, excluded and conflicting source records where material.
- Recalculate or invalidate derived evidence when a source changes or is withdrawn.

37. AI-assisted evidence

Use	Permitted status	Required safeguards
Classification/extraction	Candidate	Source link, model/version, prompt/process record, review and sampling.
Summarisation	Candidate workpaper	Traceable citations to source passages and reviewer verification.
Anomaly detection	Signal/candidate	Baseline, sensitivity, false-positive/negative review and corroboration.
Path inference	Candidate graph assertion	Approved semantics, source evidence and accountable human review.
Quality scoring	Recommendation	Transparent inputs/rules and override rationale.
Final finding/decision	Not autonomous	Competent accountable human approval required.

EVM-REQ-037 AI-generated text or graph content shall not be treated as independent corroboration of the source data from which it was generated.

Part XVI - Evidence lifecycle and review workflow

38. Lifecycle

Plan → Request → Collect → Identify → Validate → Classify → Protect → Assess Quality → Link to Assertion → Assess Sufficiency → Review Conflict/UNKNOWN → Accept or Reject → Use in Decision → Monitor Freshness → Supersede/Retain/Destroy

39. Role model

Role	Responsibility
Evidence owner	Accountable for source availability, quality and authorised use.
Collector	Acquires evidence under approved method and records provenance.
Custodian	Protects, versions, retains and controls access.
Technical specialist	Evaluates engineering or technical meaning.
Assessor	Links evidence to criteria and assertions.
Independent reviewer	Challenges sufficiency, conflict and decision use.
Decision owner	Authorises bounded conclusion and conditions.
Privacy/security/legal reviewer	Determines handling, redaction, disclosure and hold obligations.
Tool/system owner	Maintains collection and transformation integrity.

40. Evidence request record

Field	Content
requestId	Stable identifier.
decision/assertion/control	Why evidence is required.
requested type/floor	Classes, metadata, quality, period, state and format.
source/owner	Expected provider and accountable owner.
authority/purpose	Permitted collection and use.
due/review trigger	Operational scheduling field.
received/status	Collected, partial, inaccessible, rejected or UNKNOWN.
handling	Classification, transfer and retention.
substitution	Alternative evidence and approval rationale.
escalation	Decision and gate effect of non-production.

Part XVII - Evidence security, privacy and disclosure

41. Classification levels

Level	Examples	Handling intent
PUBLIC	Published methodology, synthetic examples.	Approved public release.
INTERNAL	General workpapers and non-sensitive procedures.	Authenticated organisational access.
CONFIDENTIAL	Client architecture, contracts, findings and operational records.	Need-to-know, encryption, restricted export.
RESTRICTED	Facility layouts, configurations, identities, routes and security evidence.	Strong access approval, logging and controlled sharing.
HIGHLY RESTRICTED	Combined physical/digital paths, secrets, live control authority.	Segregated storage, minimal access and sanitised reporting.
LEGAL HOLD	Any evidence under preservation obligation.	No deletion or unapproved alteration.

42. Security and privacy requirements

EVM-SEC-001 Collect only evidence necessary for the authorised assurance purpose.

EVM-SEC-002 Secrets and reusable credentials shall not be stored as ordinary evidence values.

EVM-SEC-003 Evidence access, export, redaction and deletion shall be auditable for restricted classes.

EVM-SEC-004 Personal data shall be minimised, lawfully handled and retained only as justified.

EVM-SEC-005 Cross-tenant and cross-client evidence shall remain isolated.

EVM-SEC-006 Reports may abstract sensitive details while retaining stable traceability to protected evidence.

EVM-SEC-007 Evidence used across jurisdictions shall identify location, transfer and access restrictions.

Part XVIII - Graph-native and machine-readable model

43. Core graph nodes

Node	Purpose
EvidenceObject	Addressable source item.
EvidenceSet	Assertion-specific collection.
EvidencePackage	Control/system/gate package.
EvidencePortfolio	Capability-level evidence collection.
EvidenceRequirement	Required type and threshold.
EvidenceAssessment	Quality and sufficiency judgment.
EvidenceSource	Originating system/person/provider.
CollectionActivity	Acquisition event.
TransformationActivity	Derived evidence process.
CustodyEvent	Access/transfer/change event.
Assertion	Proposition supported/challenged.
UnknownRecord	Missing or unresolved evidence condition.
EvidenceConflict	Contradictory evidence relationship.
ReviewRecord	Human challenge/approval.
Decision	Bounded assurance output.

44. Core graph relationships

Predicate	Meaning
SUPPORTS	Evidence supports a precise assertion.
CHALLENGES	Evidence weakens or contradicts assertion.
QUALIFIES	Evidence limits scope/conditions.
SATISFIES_REQUIREMENT	Evidence package meets requirement.
COLLECTED_BY	Object acquired by agent/tool.
GENERATED_BY	Object produced by activity.
DERIVED_FROM	Derived object traces to source.
OBSERVES	Evidence records an entity/state.
COVERS	Evidence covers scope, state, period or population.
VALIDATES	Test evidence evaluates a safeguard or claim.
CONFLICTS_WITH	Objects/assertions materially disagree.
SUPERSEDES	New object replaces current use.
MEMBER_OF	Object belongs to set/package/portfolio.
CUSTODIAN_OF	Party holds custody.
LIMITS	UNKNOWN/limitation constrains conclusion.
EVIDENCES_GATE	Package contributes to gate result.

45. Publication package

File	Purpose
aidcf-evm-v1.0.owl	Formal evidence ontology.
aidcf-evm-v1.0.ttl	RDF/Turtle serialisation.
aidcf-evm-context-v1.0.jsonld	JSON-LD context.
evidence-object-schema-v1.0.json	EvidenceObject exchange schema.
evidence-package-schema-v1.0.json	Package/portfolio schema.
aidcf-evm-shapes-v1.0.ttl	SHACL validation shapes.
evidence-types-v1.0.csv	Type registry.
quality-ratings-v1.0.csv	Quality dimension anchors.
control-evidence-profiles-v1.0.csv	AIDCF-007 mapping matrix.
critical-gate-evidence-v1.0.csv	Gate profile matrix.
retention-classes-v1.0.csv	Retention definitions.
conformance-tests/	Positive, negative, migration and security fixtures.
CHANGELOG.md / MIGRATION.md	Version history and transition guidance.

Part XIX - Conformance, validation and nonconformance

46. Validation rules

Rule	Validation statement
EVM-VAL-001	Every EvidenceObject has stable ID, class, source, scope, observation time, classification and review state.
EVM-VAL-002	Every accepted material EvidenceObject links to at least one assertion or requirement.
EVM-VAL-003	Every derived object links to sources and transformation activity.
EVM-VAL-004	Every quality rating has rationale and reviewer.
EVM-VAL-005	Every evidence package evaluates freshness, coverage, conflict and sufficiency.
EVM-VAL-006	Every critical-gate result references its evidence requirement and package.
EVM-VAL-007	Every missing mandatory item creates UNKNOWN or explicit adverse result.
EVM-VAL-008	No UNKNOWN is encoded as favourable pass or numeric zero.
EVM-VAL-009	No evidence item is treated as universal proof across unreviewed assertions.
EVM-VAL-010	Every stale/superseded object remains historically traceable.
EVM-VAL-011	Every restricted item has handling and access metadata.
EVM-VAL-012	Every AI-generated assertion remains candidate until approved.
EVM-VAL-013	Every evidence deletion has authorised destruction record unless prohibited by hold.
EVM-VAL-014	Every test evidence object records initial state, method, conditions and anomalies.
EVM-VAL-015	Every assurance decision links to an immutable evidence manifest or snapshot.

47. Evidence nonconformance classes

Class	Examples	Decision treatment
Critical	Fabricated/falsified evidence; concealed material conflict; unsafe unauthorised collection; custody breach affecting critical claim.	Stop use, preserve records, escalate, independently investigate and suspend affected decision.
Major	Missing provenance; unsupported effectiveness; stale critical evidence; incomplete gate package; hidden UNKNOWN.	Reject/limit evidence and reassess conclusion.
Minor	Non-material metadata defect or formatting inconsistency.	Correct and record.
Observation	Improvement opportunity without current decision impact.	Track through quality process.

Part XX - Worked evidence packages

48. Power continuity package

Component	Example
Assertions	Required power continuity and resilient capacity exist for declared load/states.
E1	SLD, design basis, protection/capacity study, switching and fuel procedures.
E2	Protection settings, UPS/generator/EPMS configuration, asset inventory.
E3	Load, runtime, fuel, alarms, maintenance and incident evidence.
E4	Generator start/transfer, UPS runtime, load-step and restoration tests.
E5	Independent engineering/witness review where required.
Quality focus	Current load/configuration, representative states, shared dependencies, test safety.
Gate	CG-05; evidence floor S5 or profile-defined strong floor.
UNKNOWN example	Unverified shared bus or fuel replenishment autonomy.
Decision limit	No resilience conclusion beyond tested load, duration and states.

49. Liquid cooling package

Component	Example
Assertions	Heat removal remains within envelope under normal and defined degraded states.
E1	P&ID, thermal model, design limits, water/coolant specification.
E2	CDU, controller, setpoint, sensor and loop configuration.
E3	Flow, temperature, pressure, leak, quality and alarm telemetry.
E4	Failover, isolation, leak response and thermal recovery test.
E5	Specialist engineering review for high-consequence or novel architecture.
Quality focus	Sensor trust, seasonal condition, transient response and shared loop dependencies.
Gate	CG-06.
UNKNOWN example	Secondary-loop autonomy not verified at peak load.
Decision limit	No generalisation from steady low load to full transient AI demand.

50. Compute-storage-network package

Component	Example
Assertions	Required usable performance, integrity and recovery are sustained for defined service outcomes.
E1	Cluster/fabric/storage architecture and compatibility basis.
E2	Firmware, driver, topology, scheduler and storage configurations.
E3	GPU errors, fabric congestion, storage performance, capacity and incident records.
E4	Benchmark, node/fabric failure, checkpoint and restore tests.
E5	Independent validation where required.
Quality focus	Representative workload, topology version, error gaps and failure-domain coverage.

Component	Example
Gate	CG-08 and CG-10.
UNKNOWN example	Checkpoint recovery not tested after correlated metadata/fabric disruption.
Decision limit	Performance evidence does not prove recovery or integrity unless directly assessed.

51. Digital authority over infrastructure

Component	Example
Assertions	Only approved identities can cause consequential infrastructure actions within stated conditions.
E1	Access policy, responsibility and remote-support process.
E2	Identity, role, permission, gateway and controller configuration.
E3	Authentication, session, command, change and revocation logs.
E4	Authorised access-path, command restriction and revocation tests.
E7	Integrated authority-path analysis using Artifact 005 semantics.
Quality focus	Effective identity, inherited privilege, dormant routes, provider access and rollback.
Gate	CG-15.
UNKNOWN example	Remote OEM command scope cannot be independently verified.
Decision limit	Reachability alone does not prove authority or realised physical effect.

Appendix A - Evidence type registry

Subtype ID	Class	Preferred label	Rule
E1-01	E1	Policy	Controlled subtype; profile may add narrower terms without redefining parent class.
E1-02	E1	Standard	Controlled subtype; profile may add narrower terms without redefining parent class.
E1-03	E1	Procedure	Controlled subtype; profile may add narrower terms without redefining parent class.
E1-04	E1	Runbook	Controlled subtype; profile may add narrower terms without redefining parent class.
E1-05	E1	Design Basis	Controlled subtype; profile may add narrower terms without redefining parent class.
E1-06	E1	Architecture Diagram	Controlled subtype; profile may add narrower terms without redefining parent class.
E1-07	E1	Single-Line Diagram	Controlled subtype; profile may add narrower terms without redefining parent class.
E1-08	E1	P&ID	Controlled subtype; profile may add narrower terms without redefining parent class.
E1-09	E1	As-Built Drawing	Controlled subtype; profile may add narrower terms without redefining parent class.
E1-10	E1	Contract	Controlled subtype; profile may add narrower terms without redefining parent class.
E1-11	E1	SLA	Controlled subtype; profile may add narrower terms without redefining parent class.
E1-12	E1	Attestation	Controlled subtype; profile may add narrower terms without redefining parent class.
E1-13	E1	Inventory	Controlled subtype; profile may add narrower terms without redefining parent class.
E1-14	E1	Risk Record	Controlled subtype; profile may add narrower terms without redefining parent class.
E2-01	E2	Device Configuration	Controlled subtype; profile may add narrower terms without redefining parent class.
E2-02	E2	Protection Setting	Controlled subtype; profile may add narrower terms without redefining parent class.
E2-03	E2	Firmware Baseline	Controlled subtype; profile may add narrower terms without redefining parent class.
E2-04	E2	Driver Matrix	Controlled subtype; profile may add narrower terms without redefining parent class.
E2-05	E2	Cluster Manifest	Controlled subtype; profile may add narrower terms without redefining parent class.
E2-06	E2	Scheduler Policy	Controlled subtype; profile may add narrower terms without redefining parent class.

Subtype ID	Class	Preferred label	Rule
E2-07	E2	Access Policy	Controlled subtype; profile may add narrower terms without redefining parent class.
E2-08	E2	Topology Export	Controlled subtype; profile may add narrower terms without redefining parent class.
E2-09	E2	Controller Logic	Controlled subtype; profile may add narrower terms without redefining parent class.
E2-10	E2	Configuration Snapshot	Controlled subtype; profile may add narrower terms without redefining parent class.
E2-11	E2	Golden Image	Controlled subtype; profile may add narrower terms without redefining parent class.
E3-01	E3	Telemetry Series	Controlled subtype; profile may add narrower terms without redefining parent class.
E3-02	E3	Event Log	Controlled subtype; profile may add narrower terms without redefining parent class.
E3-03	E3	Alarm Record	Controlled subtype; profile may add narrower terms without redefining parent class.
E3-04	E3	Capacity Metric	Controlled subtype; profile may add narrower terms without redefining parent class.
E3-05	E3	Performance Metric	Controlled subtype; profile may add narrower terms without redefining parent class.
E3-06	E3	Work Order	Controlled subtype; profile may add narrower terms without redefining parent class.
E3-07	E3	Maintenance Record	Controlled subtype; profile may add narrower terms without redefining parent class.
E3-08	E3	Incident Record	Controlled subtype; profile may add narrower terms without redefining parent class.
E3-09	E3	Change Record	Controlled subtype; profile may add narrower terms without redefining parent class.
E3-10	E3	Access Log	Controlled subtype; profile may add narrower terms without redefining parent class.
E3-11	E3	Video/Photo Observation	Controlled subtype; profile may add narrower terms without redefining parent class.
E3-12	E3	Inspection Record	Controlled subtype; profile may add narrower terms without redefining parent class.
E4-01	E4	Commissioning Test	Controlled subtype; profile may add narrower terms without redefining parent class.
E4-02	E4	Functional Test	Controlled subtype; profile may add narrower terms without redefining parent class.
E4-03	E4	Load Test	Controlled subtype; profile may add narrower terms without redefining parent class.
E4-04	E4	Failover Test	Controlled subtype; profile may add narrower terms without redefining parent class.

Subtype ID	Class	Preferred label	Rule
E4-05	E4	Recovery Test	Controlled subtype; profile may add narrower terms without redefining parent class.
E4-06	E4	Scenario Exercise	Controlled subtype; profile may add narrower terms without redefining parent class.
E4-07	E4	Simulation	Controlled subtype; profile may add narrower terms without redefining parent class.
E4-08	E4	Calibration Result	Controlled subtype; profile may add narrower terms without redefining parent class.
E4-09	E4	Witnessed Demonstration	Controlled subtype; profile may add narrower terms without redefining parent class.
E4-10	E4	Penetration/Control Test when authorised	Controlled subtype; profile may add narrower terms without redefining parent class.
E5-01	E5	Independent Engineering Review	Controlled subtype; profile may add narrower terms without redefining parent class.
E5-02	E5	Third-Party Assessment	Controlled subtype; profile may add narrower terms without redefining parent class.
E5-03	E5	Audit Report	Controlled subtype; profile may add narrower terms without redefining parent class.
E5-04	E5	Accredited Certificate	Controlled subtype; profile may add narrower terms without redefining parent class.
E5-05	E5	Laboratory Result	Controlled subtype; profile may add narrower terms without redefining parent class.
E5-06	E5	Provider Assurance Report	Controlled subtype; profile may add narrower terms without redefining parent class.
E5-07	E5	Regulatory Inspection	Controlled subtype; profile may add narrower terms without redefining parent class.
E5-08	E5	Independent Witness Record	Controlled subtype; profile may add narrower terms without redefining parent class.
E6-01	E6	Interview Record	Controlled subtype; profile may add narrower terms without redefining parent class.
E6-02	E6	Workshop Record	Controlled subtype; profile may add narrower terms without redefining parent class.
E6-03	E6	Walkthrough	Controlled subtype; profile may add narrower terms without redefining parent class.
E6-04	E6	Operator Demonstration	Controlled subtype; profile may add narrower terms without redefining parent class.
E6-05	E6	Expert Judgment	Controlled subtype; profile may add narrower terms without redefining parent class.
E6-06	E6	Shift Observation	Controlled subtype; profile may add narrower terms without redefining parent class.
E6-07	E6	Competency Record	Controlled subtype; profile may add narrower terms without redefining parent class.

Subtype ID	Class	Preferred label	Rule
E6-08	E6	Training Record	Controlled subtype; profile may add narrower terms without redefining parent class.
E7-01	E7	Trend Analysis	Controlled subtype; profile may add narrower terms without redefining parent class.
E7-02	E7	Correlation Analysis	Controlled subtype; profile may add narrower terms without redefining parent class.
E7-03	E7	Capacity Model	Controlled subtype; profile may add narrower terms without redefining parent class.
E7-04	E7	Failure Analysis	Controlled subtype; profile may add narrower terms without redefining parent class.
E7-05	E7	Dependency Graph	Controlled subtype; profile may add narrower terms without redefining parent class.
E7-06	E7	Reconciled Inventory	Controlled subtype; profile may add narrower terms without redefining parent class.
E7-07	E7	Evidence Quality Assessment	Controlled subtype; profile may add narrower terms without redefining parent class.
E7-08	E7	Score/Rating Output	Controlled subtype; profile may add narrower terms without redefining parent class.
E7-09	E7	AI-Assisted Extraction	Controlled subtype; profile may add narrower terms without redefining parent class.
E7-10	E7	Query Result	Controlled subtype; profile may add narrower terms without redefining parent class.

Appendix B - Quality scoring anchors

P

Rating	Anchor
P0	Unknown origin
P1	Identified but uncontrolled source
P2	Accountable internal source
P3	Controlled authoritative source
P4	Corroborated authoritative source
P5	Independently verified origin and lineage

F

Rating	Anchor
F0	Unknown age or state
F1	Obsolete for represented state
F2	Stale or change-sensitive
F3	Acceptable for decision
F4	Current and aligned to baseline

Rating	Anchor
F5	Near-real-time or event-current with healthy collection

C

Rating	Anchor
C0	Unknown coverage
C1	Anecdotal or isolated
C2	Limited sample/state
C3	Representative core scope
C4	Broad scope and relevant states
C5	Comprehensive declared population, period and states

R

Rating	Anchor
R0	Unassessed
R1	Weak/uncontrolled method
R2	Plausible but lightly corroborated
R3	Controlled repeatable method
R4	Strong and independently corroborated
R5	Validated method and independent verification

T

Rating	Anchor
T0	Unlinked
T1	Manual contextual link
T2	Linked to subject or control
T3	Linked to assertion, subject and period
T4	End-to-end to finding and gate
T5	End-to-end to decision with immutable lineage

I

Rating	Anchor
I0	Unknown integrity
I1	Basic repository controls
I2	Verified source copy
I3	Hash verified
I4	Signature/timestamp verified
I5	Tamper-evident end-to-end custody

Appendix C - Control evidence profile matrix

The 360-control evidence profile matrix is maintained in the AIDCF-007 companion catalogue and shall be published as control-evidence-profiles-v1.0.csv.

Appendix D - Critical-gate evidence matrix

Gate	Name	Mandatory evidence	Classes	Floor	Decision effect
CG-01	Capability and outcome definition	Capability register; approved outcomes; criticality and tolerance analysis; architecture linkage	E1+E7	S3	UNKNOWN capability/outcome blocks positive capability assurance.
CG-02	Boundary and architecture integrity	Approved boundary; current architecture; as-built/configuration reconciliation; exclusions and external dependencies	E1+E2+E6	S3	Unclear material boundary blocks final conclusion.
CG-03	Critical dependency visibility	Dependency graph; source/route/provider evidence; common-mode analysis; ownership	E1+E2+E7	S4	Critical hidden dependency remains UNKNOWN or Fail if demonstrated absent.
CG-04	Life safety and hazardous energy	Authorised engineering/safety records; inspection; permits; tests; unresolved hazard status	E1+E4+E5	S4	Unsafe unresolved condition blocks relevant assurance/testing.
CG-05	Resilient electrical supply	SLD; protection/capacity study; UPS/generator tests; fuel; maintenance; EPMS/incident evidence	E1+E2+E3+E4	S5	Missing critical continuity evidence → UNKNOWN; demonstrated inability → Fail.
CG-06	Thermal continuity	Thermal design/model; P&ID; capacity/flow telemetry; leak/transfer/failover tests; maintenance	E1+E2+E3+E4	S5	Unverified critical heat-removal path → UNKNOWN; demonstrated shortfall → Fail.
CG-07	Critical control-system integrity	BMS/EPMS/controller architecture; configuration; identity/authority; telemetry; fail-safe tests	E1+E2+E3+E4	S4	Uncontrolled or unverifiable critical automation limits/blocks assurance.
CG-08	Compute, storage and network functional integrity	Current topology/configuration; capacity/performance telemetry; error data; failover/recovery tests	E2+E3+E4	S4	Unverified critical infrastructure function caps operating/resilience assurance.
CG-09	Failure containment	Fault-domain model; isolation design; scenario analysis; containment tests/incidents	E1+E2+E4+E7	S4	Absent/failed containment may be Fail; untested critical boundary is UNKNOWN.
CG-10	Recovery readiness	Recovery plan; prerequisites; configuration/data sources; exercises/tests; defects and signoff	E1+E2+E3+E4	S5	Plan without representative test supports design only, not recoverability.
CG-11	Evidence sufficiency	Evidence requirements; manifest; quality/sufficiency assessments; conflicts and limitations	E1+E7	S4	Failure to meet evidence floor blocks the dependent claim.
CG-12	Critical UNKNOWN	UNKNOWN register; consequence; resolution ownership; evidence requests; decision ceiling	E1+E7	S3	Unresolved critical UNKNOWN caps, conditions, suspends or blocks assurance.
CG-13	Responsibility and authority	RACI; delegations; owner/operator/provider clauses; escalation; decision authority	E1+E6	S3	Unowned critical function/dependency blocks accountable disposition.
CG-14	Change control	Baseline; change request; impact analysis; test/rollback; approval; post-change validation	E1+E2+E3+E4	S4	Material uncontrolled change suspends affected evidence/decision.
CG-15	Integrated digital authority	Identity and privilege records; reachability; control-plane interfaces; session logs; action/rollback tests	E2+E3+E4+E7	S4	Unverified consequential authority remains critical UNKNOWN.

Appendix E - Record schemas

E.1 EvidenceObject

Property	Cardinality	Type
evidenceId	1	Controlled value or referenced object
title	1	Controlled value or referenced object
description	0..1 / 0..*	Controlled value or referenced object
evidenceClass	1	Controlled value or referenced object
subtype	0..1 / 0..*	Controlled value or referenced object
sourceId	1	Controlled value or referenced object
owner	0..1 / 0..*	Controlled value or referenced object
collector	0..1 / 0..*	Controlled value or referenced object
collectionMethod	0..1 / 0..*	Controlled value or referenced object
observationTime	1	Controlled value or referenced object
collectionTime	0..1 / 0..*	Controlled value or referenced object
effectivePeriod	0..1 / 0..*	Controlled value or referenced object
subjectRefs	0..1 / 0..*	Controlled value or referenced object
scope	0..1 / 0..*	Controlled value or referenced object
configurationVersion	0..1 / 0..*	Controlled value or referenced object
integrityMethod	0..1 / 0..*	Controlled value or referenced object
hash	0..1 / 0..*	Controlled value or referenced object
classification	1	Controlled value or referenced object
retentionRule	0..1 / 0..*	Controlled value or referenced object
qualityAssessment	0..1 / 0..*	Controlled value or referenced object
reviewState	1	Controlled value or referenced object
limitations	0..1 / 0..*	Controlled value or referenced object
relatedEvidence	0..1 / 0..*	Controlled value or referenced object

E.2 EvidenceAssessment

Property	Meaning
assessmentId	Stable ID.
assertionOrRequirement	Precise evaluation target.
evidenceRefs	Items assessed.
qualityVector	P/F/C/R/T/I ratings and rationale.
sufficiency	S0-S5 and rationale.
confidence	Controlled assertion confidence.
conflictsAndUnknowns	Unresolved conditions.
reviewer/competence	Accountable reviewer.
date/snapshot	Temporal and version context.
decisionEffect	Finding, gate or claim ceiling.

E.3 EvidenceManifest

Property	Meaning
manifestId/version	Immutable package identity.
package/portfolio	Parent scope.
evidenceItems	IDs, hashes, versions, locations and states.
missingItems	Expected but absent objects.
restrictedItems	Protected references and access rules.
supersession	Replaced objects and lineage.
signature/timestamp	Manifest integrity method.
review/approval	Release roles and date.

Appendix F - Review and release gates

Review role	Status	Focus
Methodology author	Draft completed	Evidence architecture coherence.
Assurance/evidence specialist	Pending	Quality, sufficiency and confidence model.
Electrical engineering	Pending	CG-05 evidence profile.
Mechanical/liquid cooling	Pending	CG-06 evidence profile.
Compute/storage/network	Pending	CG-08/10 evidence profile.
Operations/reliability	Pending	Operational and recovery evidence.
OT/cybersecurity	Pending	Control-plane, identity and logging evidence.
Ontology/graph	Pending	Evidence classes/relationships and Artifact 005 alignment.
Data/privacy/security	Pending	Classification, minimisation and handling.
Legal/records management	Pending	Retention, legal hold and disclosure.
Certification/accreditation	Pending	Evidence floors and scheme implications.
Independent practitioner	Pending	Usability and reproducibility.
Public release	Not approved	Mandatory review gates closed.

Acceptance criteria

- All AIDCF-007 controls can receive a complete Evidence Requirement Profile.
- Evidence quality dimensions remain visible and non-compensating where claim-critical.
- UNKNOWN, conflicts and stale evidence cannot be hidden by scoring.
- Critical-gate evidence floors are enforceable.
- Human and machine-readable records preserve provenance and review.
- Security, privacy, retention and lawful handling are operationally usable.
- Artifact 009 can directly consume the model for assessment procedures.

Change record

Version	Status	Summary
0.1	Architecture draft	Evidence doctrine, hierarchy and taxonomy.
0.5	Integrated draft	Quality, sufficiency, gate, graph and lifecycle models.
1.0	Controlled-review draft	Full evidence standard and 360-control profile matrix.

Final doctrine

Evidence is the currency of AIDCF assurance. An evidence item has no universal authority outside its provenance, scope, time, state, method and assertion. A strong-looking package cannot compensate for a missing mandatory type, critical coverage gap, unresolved conflict or failed gate. When evidence cannot support the requested conclusion, AIDCF preserves UNKNOWN, states the limitation and withholds stronger assurance rather than manufacturing certainty.